OLED 실험 5주차

2026학년도 2학기

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Organic Electronic Materials Laboratory

주 (이론)	실험 담당	실험명
1주차	조교 전원	OT
2주차	조승현	실험 1. UV-Vis spectrum 측정에 따른 변화 분석 **Optical band gap ?**
3주차	조승현	실험 2. 형광, 인광 측정
4주차		추석 (휴강)
5주차	이채민	실험 3. 분자 시뮬레이션을 통한 HOMO/LUMO 전자 구름 분포 분석
6주차	조승현	실험 4. Solution 박막 제작 및 박막과 용액의 흡수 및 PL 비교
7주차	조승현	실험 5. CV 측정 **HOMO, LUMO ?**

5주차: Cyclic voltammetry를 이용한 HOMO 준위 측정 (조승현)

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- Cyclic voltammetry(CV)의 기본 원리와 3전극 전기화학 측정 시스템의 구성을 이해한다.
- Ferrocene을 기준 물질로 사용하여 유기물의 산화 개시 전위(oxidation onset potential)를 측정하고, 이를 이용해 HOMO 에너지 준위를 계산한다.
- Scan rate 변화에 따른 cyclic voltammogram의 변화를 비교하여 전극 반응 및 물질 전달의 영향을 이해한다.

Measurement Method of HOMO Level

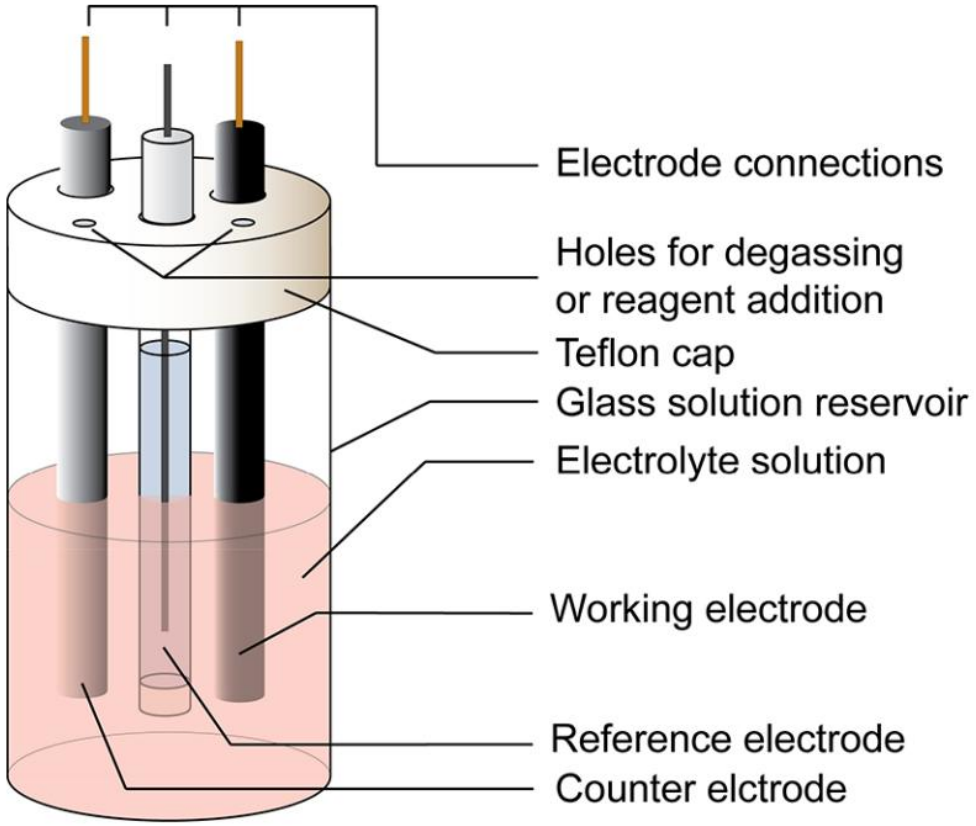
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Three-Electrode System

❖ Working Electrode (작업전극)
분석 대상 물질의 산화·환원 반응이 일어나는 전극

❖ Reference Electrode (기준전극)
일정하고 안정한 기준 전위를 제공하는 전극으로, 이상적으로는 거의 전류가 흐르지 않음. 평형상태의 금속 전위 값이나 비가역상태의 부식전위 값을 구할 때 기준이 되는 전극

❖ Counter Electrode (보조전극)
Working electrode에서 발생하는 전류가 흐를 수 있도록 회로를 완성하는 전극
일반적으로 Pt와 같은 화학적으로 안정하고 전기전도성이 높은 전극 사용



Measurement Method of HOMO Level

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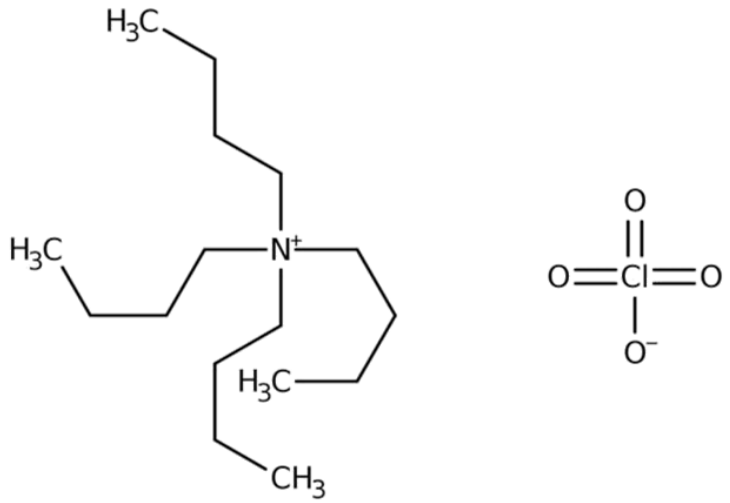
❖ Electrolyte (전해질)

용액에서 이온을 형성하여 전기전도성을 제공하는 물질

❖ Supporting Electrolyte

CV 측정에서는 충분한 농도의 supporting electrolyte를 첨가하여 용액의 전도도를 높임

solution resistance를 감소시키고 물질 전달이 주로 diffusion에 의해 일어나도록 함



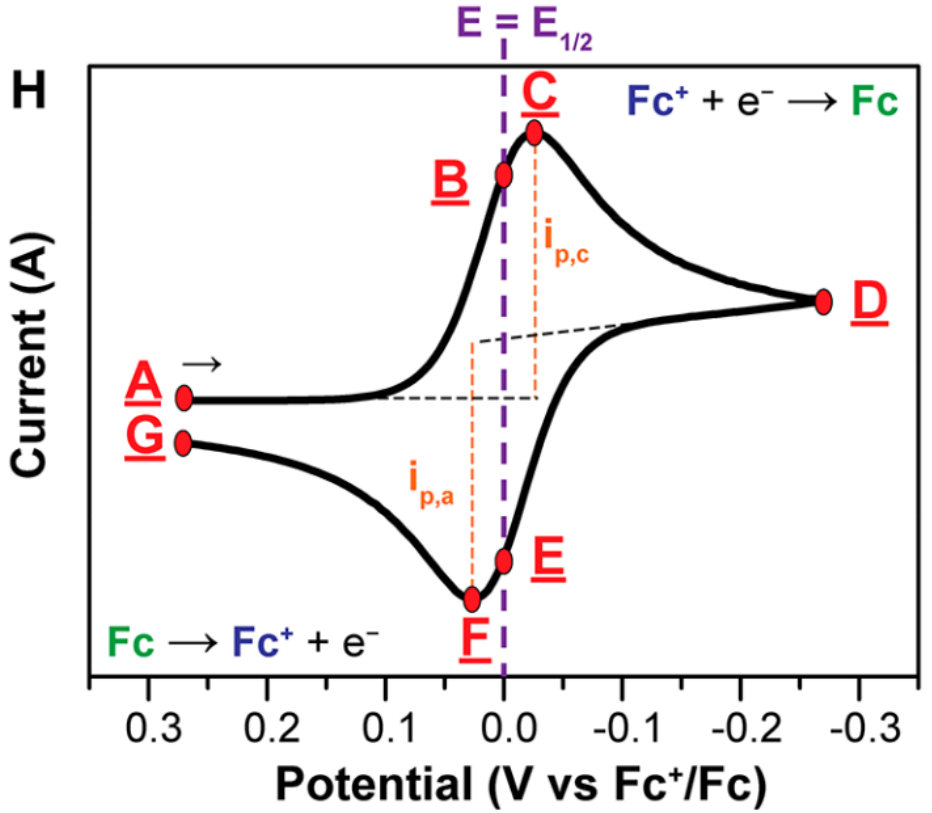
< Tetra-n-butylammonium perchlorate (TBAP) >

Measurement Method of HOMO Level

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❖ Cyclic voltammetry

- Reference standard: Ferrocene (Fc/Fc^+)
- Energy level of Fc/Fc^+:
  −4.8 eV vs. vacuum level
- Method for the calculation of HOMO level of material:
  $E_{\mathrm{HOMO}} = -4.8 - (E_{\mathrm{onset}} - E_{FC})$

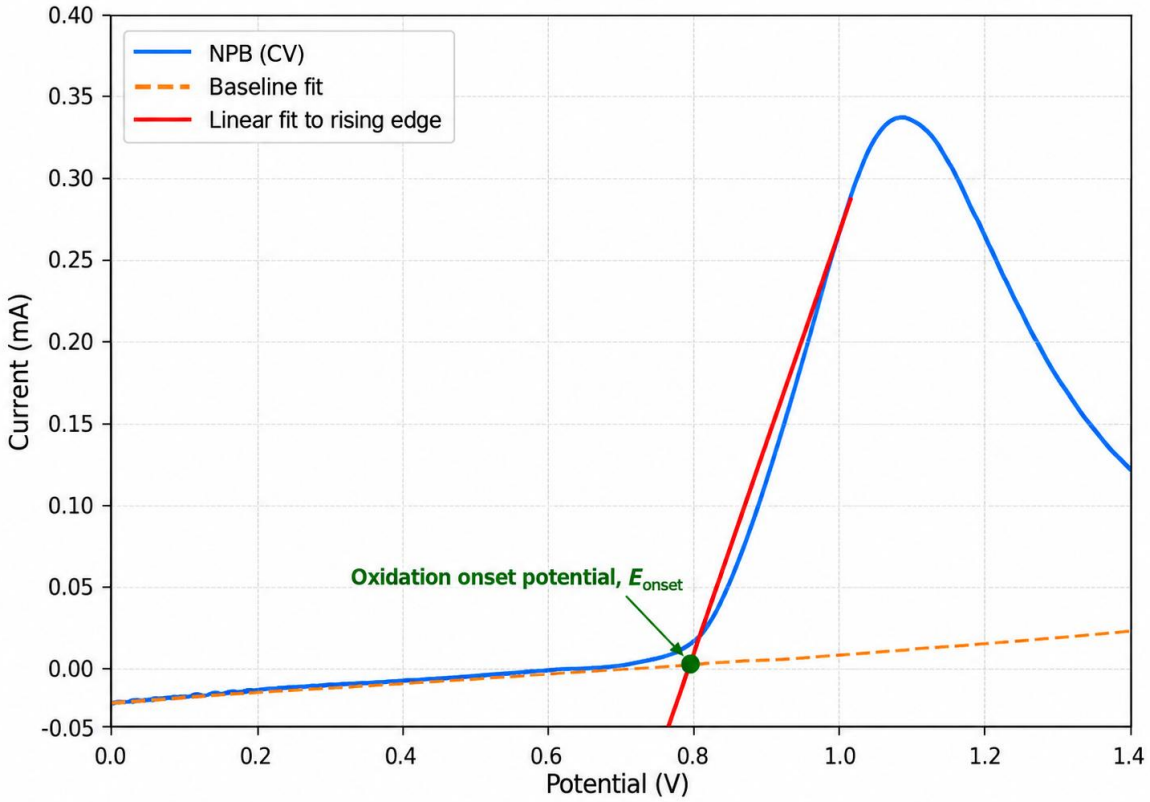


❖ Experimental setup

- Working electrode: ITO
- Counter electrode: Pt
- Reference electrode: Ag/Ag^+
- Potentiostat (전위차계) & electrochemical cell
- Supporting electrolyte: TBAP [$(n\text{-}Bu)_4NClO_4$]

$$E^0 \approx E_{1/2,FC} = \frac{E_a + E_c}{2}$$

Cyclic voltammetry



$$\mathbf{NPB} \rightarrow \mathbf{NPB}^{+} + e^{-}$$

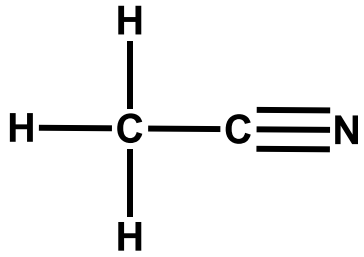
$$E_{\mathbf{HOMO}}(\mathbf{eV}) = -4.8 - (E_{\mathbf{onset,NPB}} - E_{1/2,FC})$$

- Baseline: 산화 반응이 본격적으로 시작되기 전 전류 변화가 거의 없는 구간의 기울기를 구함
- 산화 전류가 급격하게 증가하는 구간에서 접선을 긋고, 두 직선이 만나는 전위를 E_{onset} 으로 정의

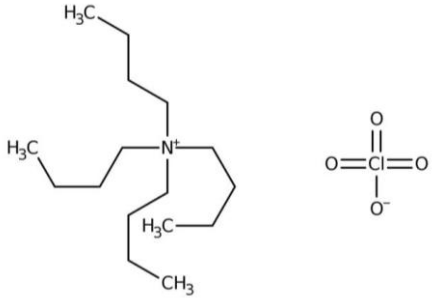
5주차: Cyclic voltammetry를 이용한 HOMO 준위 측정 (조승현)

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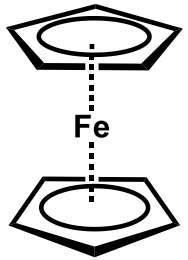
➢ 실험 재료



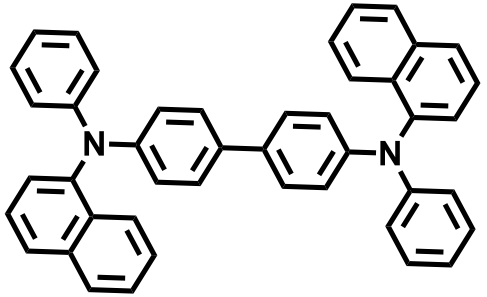
< Acetonitrile >



< Tetra-n-butylammonium perchlorate (TBAP) >



< Ferrocene >



< NPB >

➢ 실험 결과

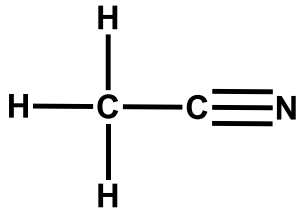
- CV를 통해 측정한 NPB의 Oxidation potential을 결정하고, Ferrocene의 HOMO level (4.8 eV)을 이용하여 측정한 물질(NPB)의 HOMO를 계산하시오.
- NPB의 UV–Vis absorption spectrum으로부터 optical band gap을 계산하시오.
  계산한 HOMO level과 optical band gap을 이용하여 LUMO level을 추정하고,
  실험을 통해 얻은 HOMO, LUMO, Band gap을 문헌값과 비교하시오.
- Scan rate에 따른 Ferrocene의 cyclic voltammetry 측정 결과를 비교하고 분석하시오.

5주차: Cyclic voltammetry를 이용한 HOMO 준위 측정 (조승현)

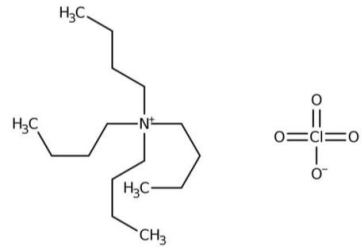
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◆ 용액 제조 – Weighing (계량)

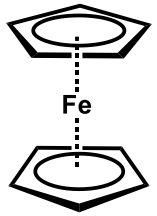




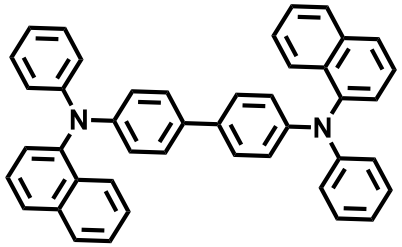
< Acetonitrile >



< TTBAP >



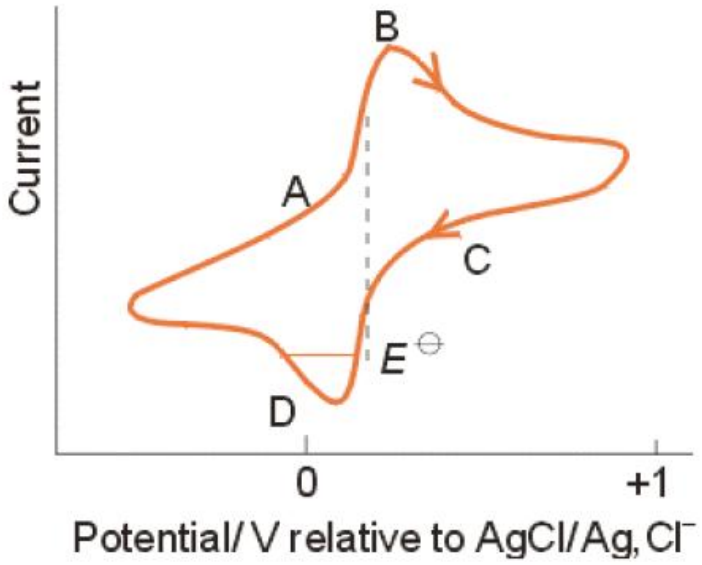
< Ferrocene >



< NPB >

Appendix

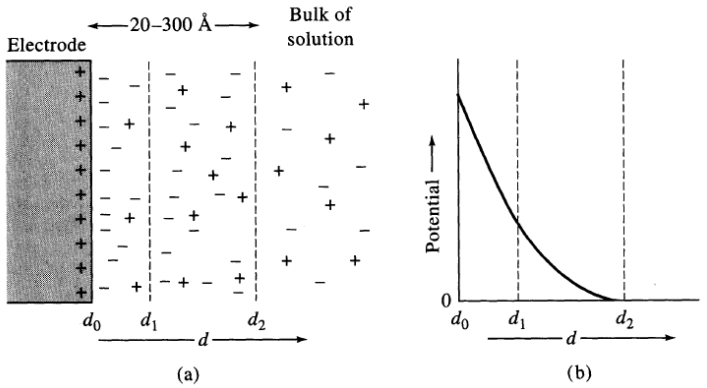
Cyclic voltammetry

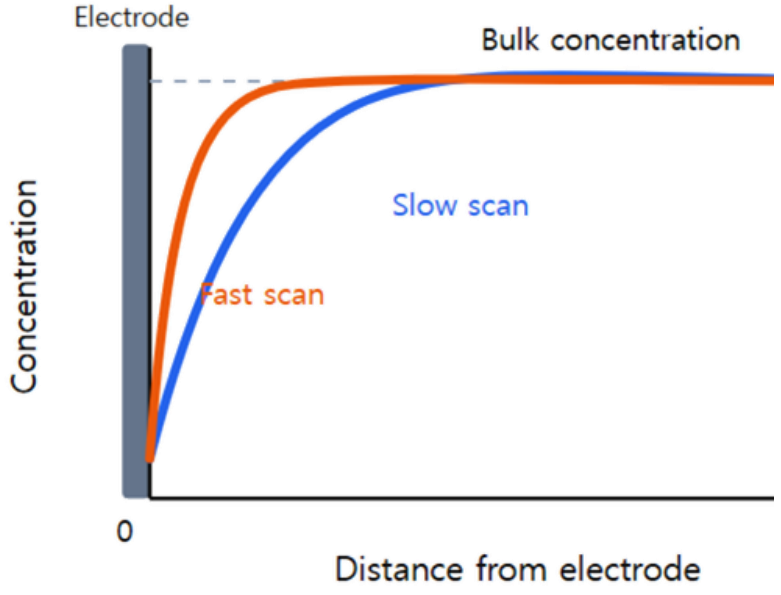


Electrical double layer formed

(i) Tightly bound inner layer

(ii) Loosely bound outer layer

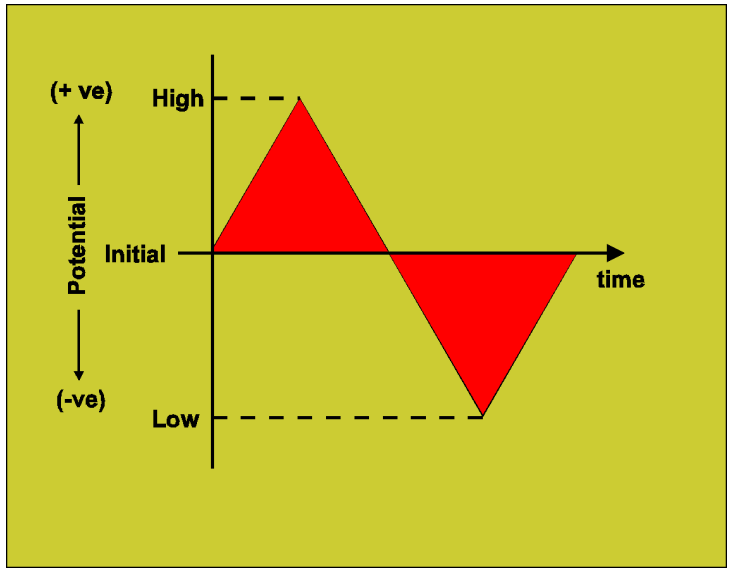


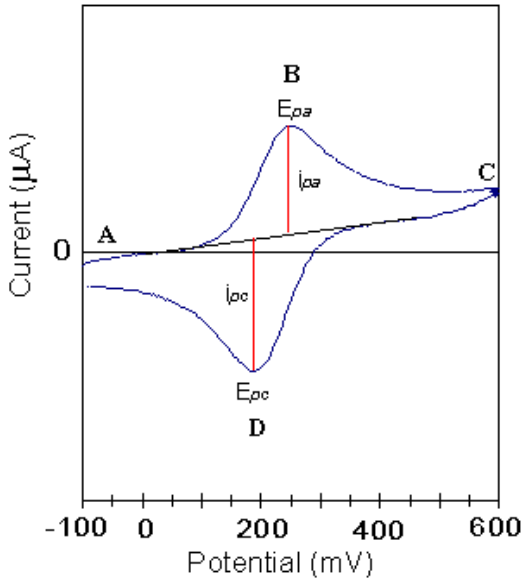


Measurement Method of HOMO Level

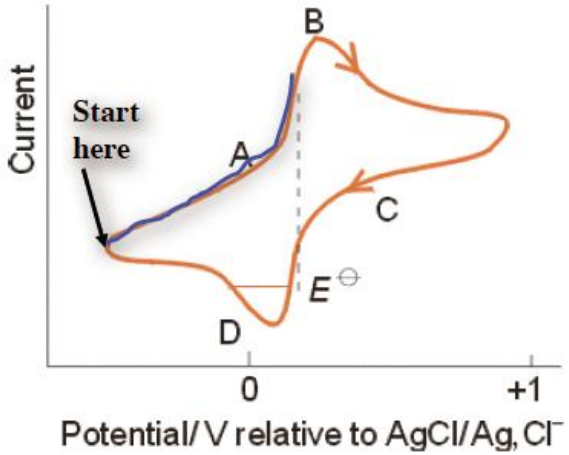
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- 시간에 따라 전위를 일정한 주사 속도(scan rate)로 변화시키면서 전극에 흐르는 전류를 측정하고, 이를 전류–전위 곡선으로 기록하는 방법을 전위 주사법(potential sweep method)이라고 한다.
- 이때 설정한 전위 범위에서 전위를 한 방향으로 주사한 뒤 다시 반대 방향으로 되돌려 측정하는 방법을 순환 전압전류법(cyclic voltammetry, CV)이라고 한다.
- CV를 이용하면 물질의 산화·환원 특성과 산화·환원 전위를 측정할 수 있다.



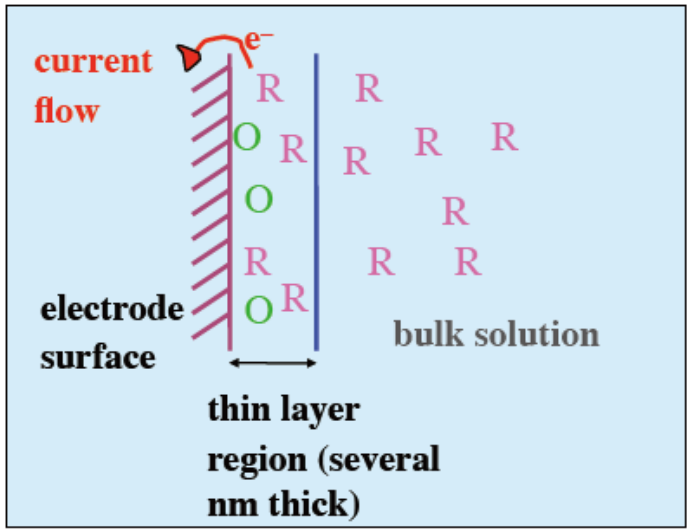


Cyclic voltammetry



$$O + e^- \rightleftharpoons R$$

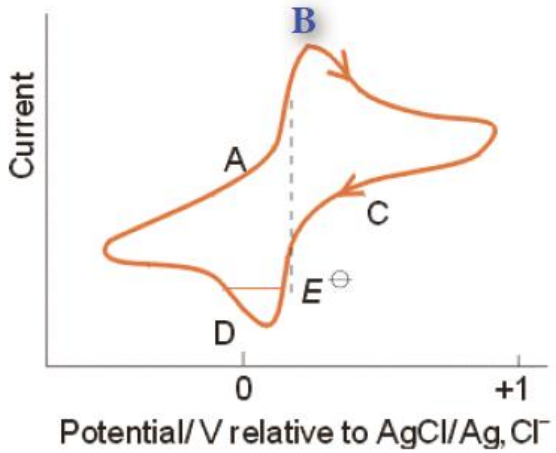
oxidized species — reduced species

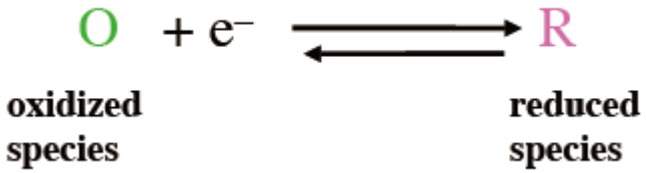


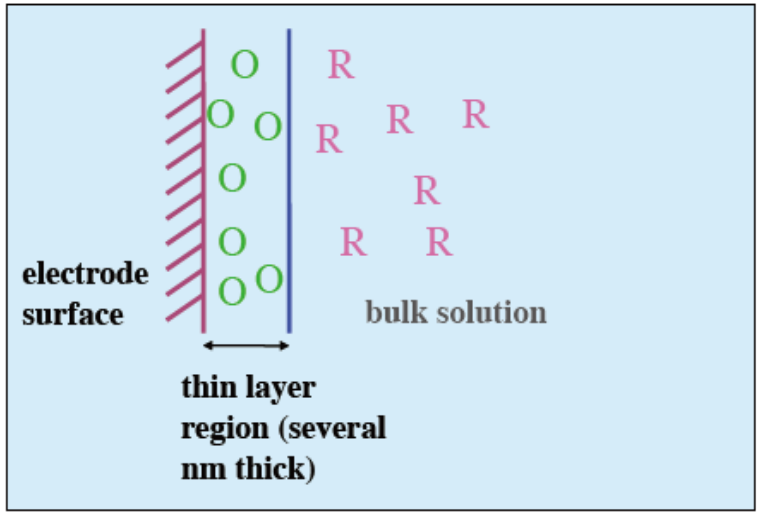
Region A:

- starting to oxidize the reduced species, R
- converting it to O the oxidized species

Cyclic voltammetry



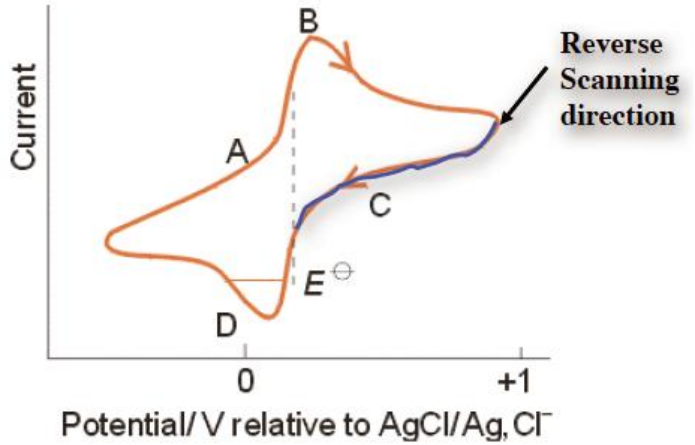




Point B:

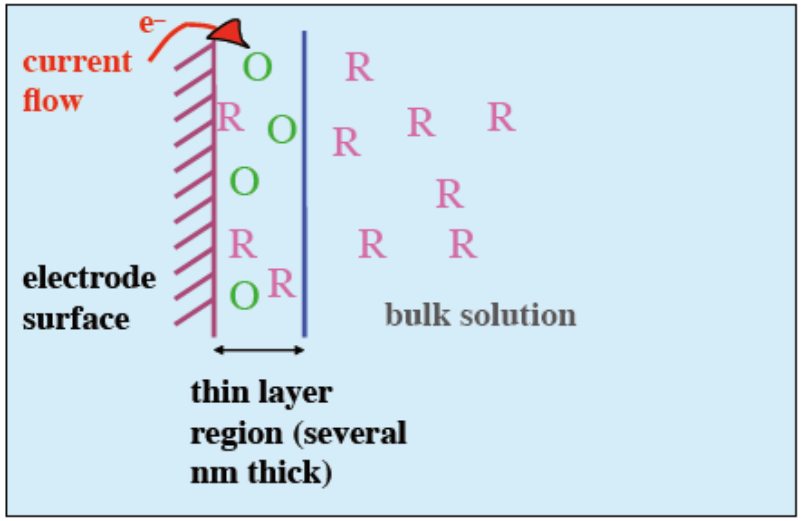
- all of the reduced species, R, in the thin layer region have been oxidized

Cyclic voltammetry



$$O + e^- \rightleftharpoons R$$

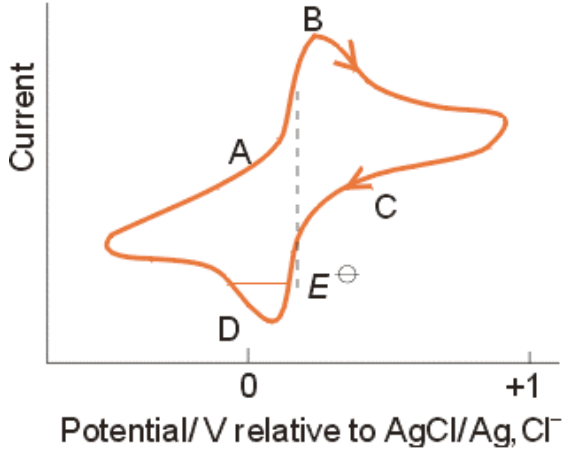
oxidized species — reduced species



Region C:

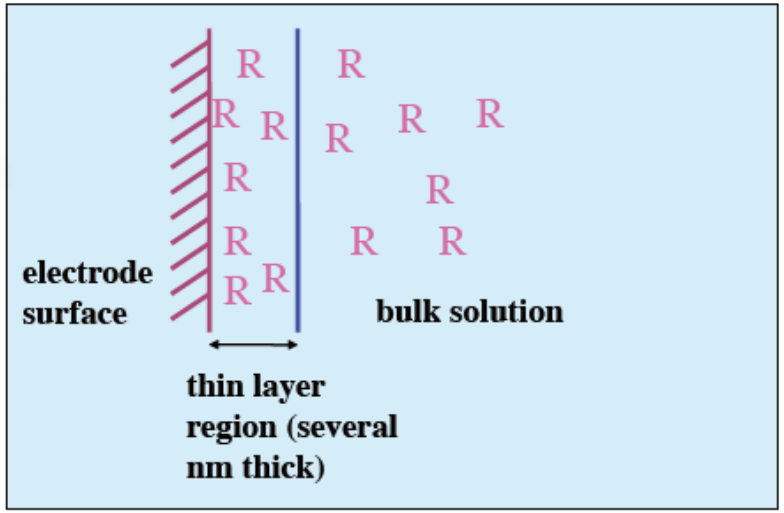
- starting to re–reduce the oxidized species, O, contained in the thin layer region

Cyclic voltammetry



$$O + e^- \rightleftharpoons R$$

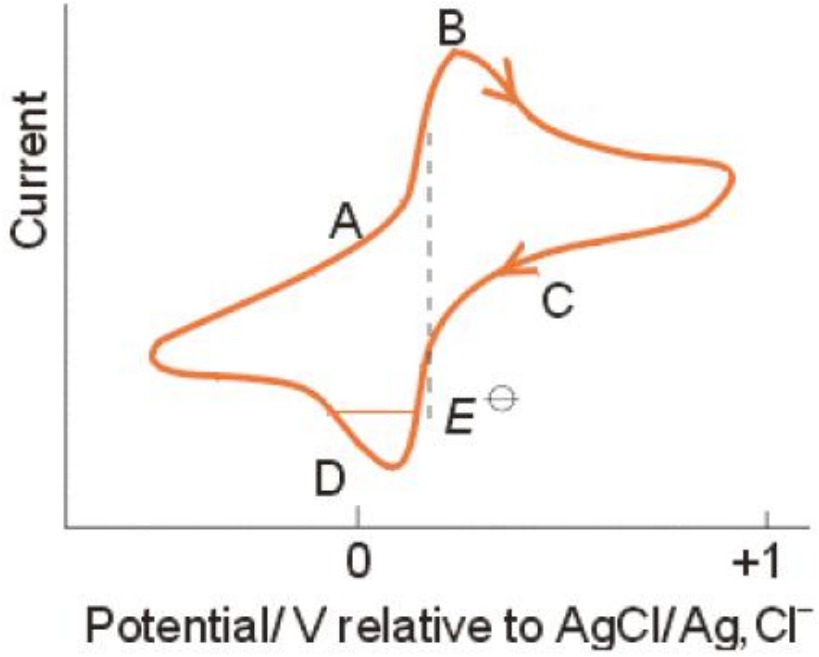
oxidized species — reduced species



Point D:

- have re–reduced all of the oxidized species, O, in the thin layer region

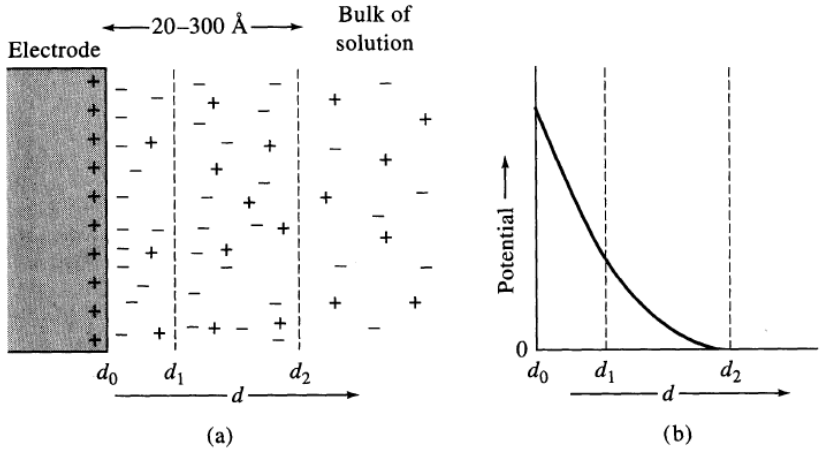
Cyclic voltammetry



Electrical double layer formed

(i) Tightly bound inner layer

(ii) Loosely bound outer layer



Current vs. potential wave associated with the reaction:

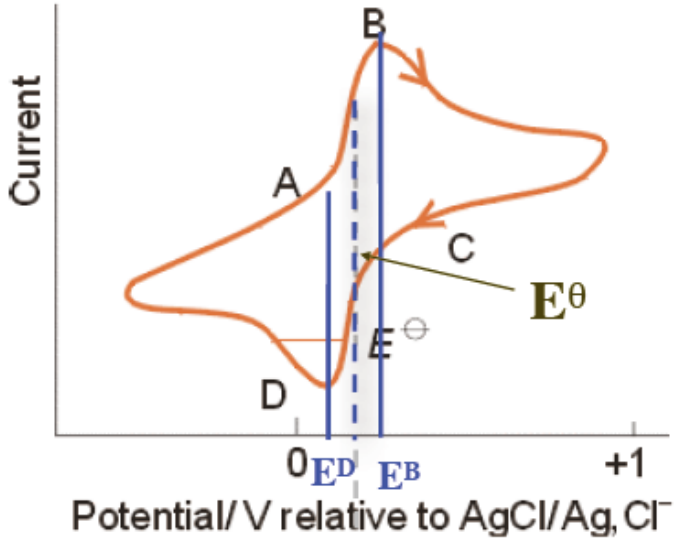
$$O + e^- \longrightarrow R$$

oxidized species **reduced species**

Cyclic voltammetry

$$O + e^- \rightleftharpoons R$$

oxidized species **reduced species**



The potential labeled E^θ :

E^θ = potential at which O is reduced

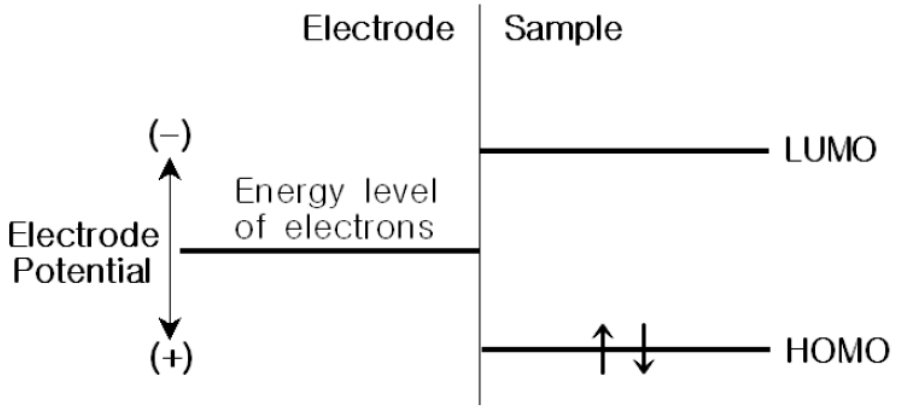
E^θ = potential at which R is oxidized

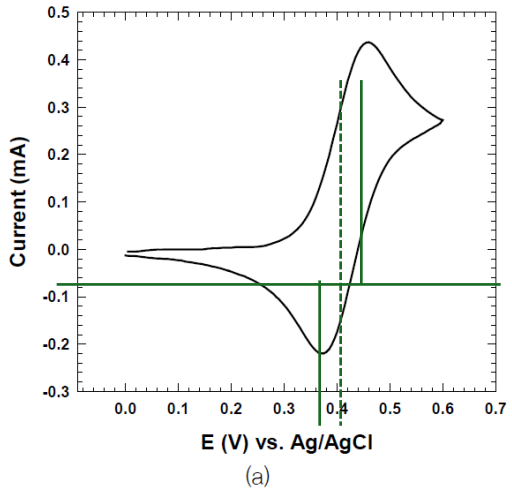
$$E^\theta = \frac{E^B + E^D}{2}$$

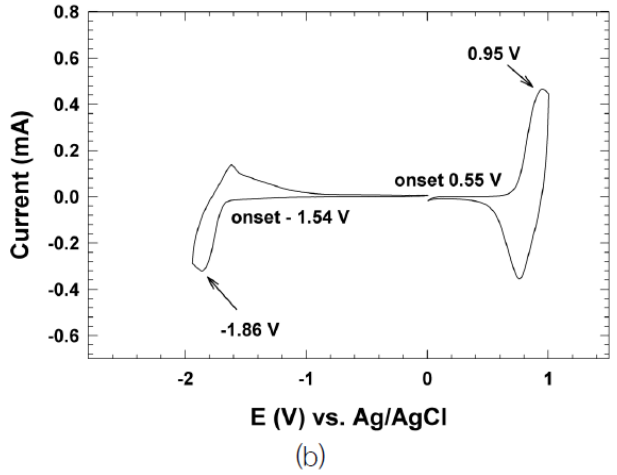
- E^θ is half–way between E^B and E^D
- thus E^θ is frequently referred to as $E_{1/2}$

Cyclic voltammetry

$$\text{HOMO (or LUMO) (eV)} = -4.8 - (E_{onset} - E_{1/2}(\text{Ferrocene}))$$







HOMO?
LUMO?

$E_{1/2}$ (ferrocene) = 0.42 V (vs. Ag/AgCl)